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Abstract

Minor motion magnification aims to reveal subtle spatial or photometric variations in videos that are difficult to perceive directly but are diagnostically and analytically meaningful in visual computing. This review examines the evolution of the field from handcrafted Lagrangian and Eulerian signal-processing pipelines to recent deep learning-based frameworks that learn motion representations,

1 Introduction

Minor motion is a fundamental phenomenon observed across multiple scales, from microscopic interactions to macroscopic systems [1, 2]. Such subtle variations are typically induced by weak signals, including mechanical vibrations, slight deformations in biological tissues, and minor environmental fluctuations, which are often difficult to perceive directly in raw images or videos [3–5].

To address this challenge, minor motion magnification (MM), commonly referred to in the literature as video motion magnification (VMM), has emerged as an effective technique for amplifying imperceptible spatio-temporal variations, making them visible and measurable.

This capability has enabled a wide range of applications. In structural health monitoring, MM facilitates early-stage fault detection [6–8]. In biomedical analysis, it supports non-contact physiological measurement and diagnostic assessment [9–13]. In industrial inspection, it enhances precision monitoring and quality control [14–17]. These developments have established minor motion analysis as an important research topic in computer vision and image processing [18, 19]. For example, MM can be used to analyze vibration modes of machinery for early fault diagnosis [20], or to estimate comparatively limited attention given to the systematic organization of deep ing-based methods. Second, deep learning approaches are often treated as mono- end-to-end models, without explicitly analyzing how different architectural onents (e.g., convolutional operations, attention mechanisms, or multi-domain sentations) contribute to motion disentanglement and noise suppression. Third, cation scenarios are typically discussed in isolation, lacking a unified cross-domain ective that connects algorithmic design with domain-specific constraints such as r noise and environmental variability [4, 24]. Furthermore, a comprehensive sum- of datasets, evaluation protocols, and benchmarking strategies remains limited, n hinders reproducibility and fair comparison.

o address these gaps, this review is structured around three key research ques- : (1) how minor motion magnification can be formulated within a unified ematical framework; (2) how deep learning-based methods evolve from classi- pproaches and differ across architectural paradigms; and (3) how these methods rm across diverse application domains under realistic conditions.

ased on this perspective, the main contributions of this paper are summarized as vs:

unified formulation of minor motion magnification is presented, together with

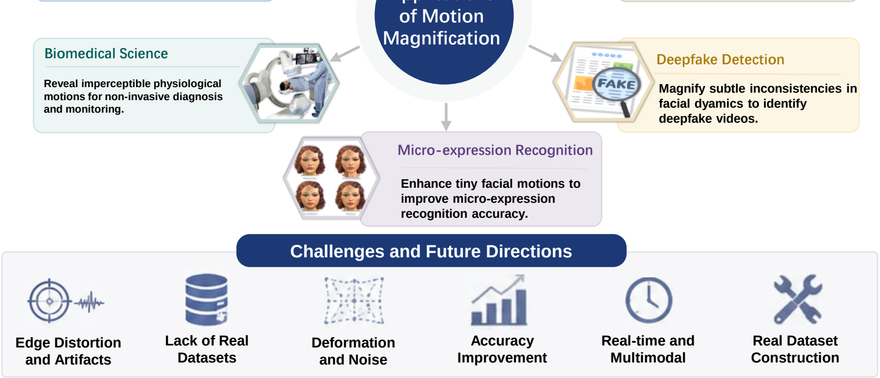
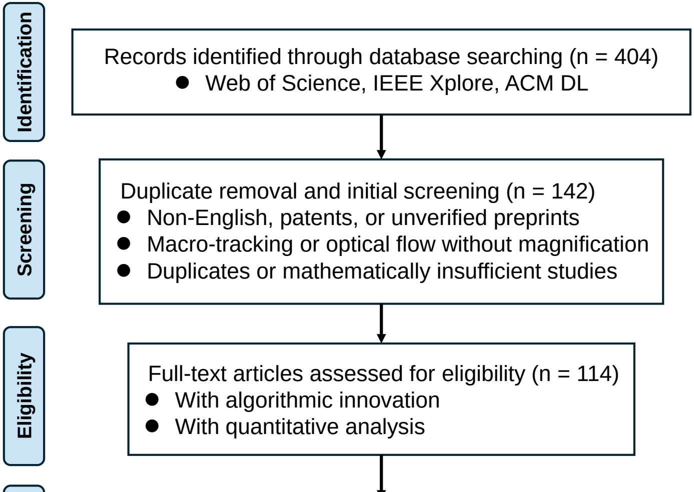


Fig. 1 Overall structure and organization of the review, illustrating the taxonomy of motion magnification methods, application domains, and research workflow.

evaluation protocols, and benchmarking strategies. Subsequently, representative application domains are discussed, including visual vibration measurement, structural health monitoring, biomedical analysis, micro-expression recognition, and deepfake detection. Finally, current challenges and future research directions are outlined.

. Ultimately, 64 representative works were retained for comprehensive analysis
comparative synthesis in the following sections.



based methods. A comparative overview of these methodologies is presented in Figure 4. As illustrated, traditional methods rely on handcrafted representations and predefined filters, whereas deep learning-based approaches learn motion representations directly from data, enabling more adaptive and scalable magnification performance across diverse scenarios.

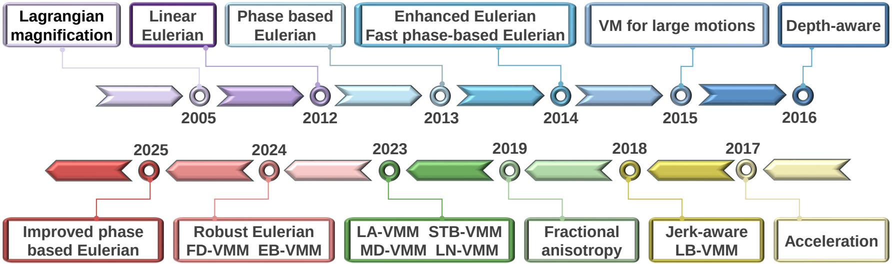
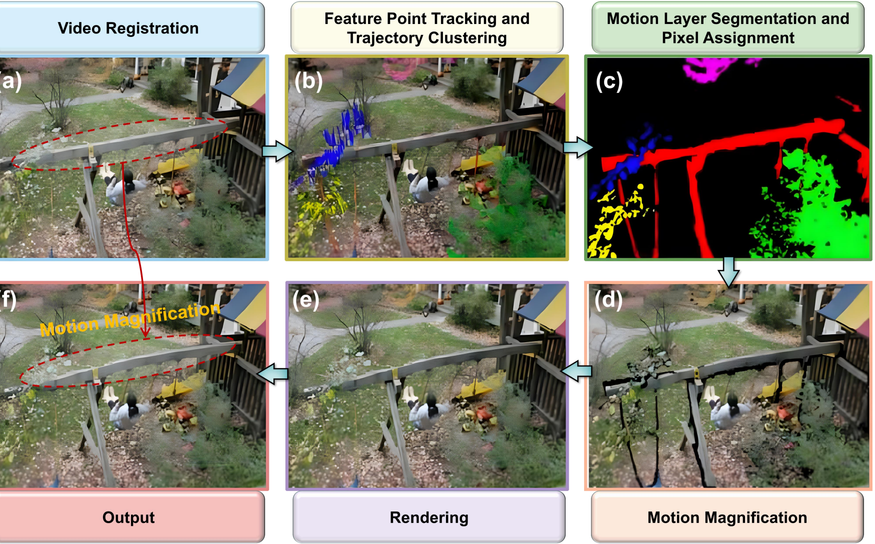


Fig. 3 Motion magnification development timeline. 2005: [25]. 2012:[23]. 2013:[26]. 2014: [27, 28]. 2015:[29]. 2016: [30]. 2017: [31]. 2018:[32, 33]. 2019: [34]. 2023:[35–39]. 2024:[40–45]. 2025:[46–48].



5 The Lagrangian motion magnification procedure[25] consists of the following steps: (a) Video
ation of the original frame to be magnified; (b) Feature point trajectory tracking and trajectory

enables a precise analysis of the dynamic behavior of the observed features.

After extracting suitable feature points, they are then grouped based on the similarity of their motion trajectories, such as speed and direction. Clustering algorithms like K-means[58] and DBSCAN[59] are commonly used for this task, as they effectively group feature points with similar dynamic behaviors. This clustering process helps to identify patterns within the motion, facilitating a more structured analysis of the video's movement characteristics.

III. Motion Layer Segmentation and Pixel Assignment

Based on the clustering results of feature point trajectories, we can further refine the processing and segmentation of pixels within the image. This involves carefully analyzing the motion trajectories of pixels within each group and merging those with similar motion patterns or paths into the same motion layer. This approach not only enhances our ability to understand and distinguish the movement states of different objects within the image but also provides robust support for subsequent tasks such as motion analysis and object tracking.

For non-feature point pixels, we can assign them to the most appropriate motion layer based on their appearance characteristics, such as color, intensity, and texture. This assignment process typically employs image segmentation optimization algorithms[60–62], which help ensure that each pixel is grouped with the most fitting opment and widespread application. However, this groundbreaking research has solid foundation for subsequent motion magnification techniques and has demoned notable potential for application in fields like structural health monitoring and cal imaging.

2 Eulerian

Eulerian is similarly inspired by fluid mechanics. Unlike the Lagrangian perive, the Eulerian perspective excels at analyzing the temporal evolution of -position pixels, making it particularly suitable for the analysis of local small ments, especially in applications such as motion magnification. Based on differrocessing techniques, Eulerian methods can be divided into linear Eulerian video ification and phase-based Eulerian video magnification.

Linear Eulerian Video Magnification

inear Eulerian Video Magnification (LEVM) was first proposed by Wu et al.[23]. LEVM does not track motion as the Lagrangian method does; instead, it amplifies on using temporal processing based on a first-order Taylor series expansion comy used in optical flow analysis[56, 66].The approach includes three main steps, al decomposition, temporal processing, reconstruction, as shown in Figure 6. Spaured for specific application scenarios.

To provide an intuitive and mathematically tractable demonstration of the relationship between temporal filtering and motion scaling, we simplify the generalized 2D formulation into a directional one-dimensional (1D) spatial signal. This simplified formulation preserves the essential characteristics of local translational motion fields. Under this 1D formulation, let $I(x,t) = f(x + \delta(t))$ denote the observed image intensity at spatial position x and time t. Here, $f(x) = I(x, 0)$ explicitly defines the static canonical reference frame, while $\delta(t)$ signifies the time-varying, subtle displacement function. The primary objective of motion magnification is to synthesize an amplified target reconstruction signal, $\hat{I}(x,t)$, wherein this underlying displacement is scaled by a user-specified amplification factor $\alpha \in \mathbb{R}^+$, expressed as:

$$\hat{I}(x,t) = f(x + (1+\alpha)\delta(t)) \tag{1}$$

This methodology is formally designated as Linear Eulerian Video Magnification (LEVM) because it operates under the assumption that the motion-induced intensity variation can be linearized with respect to the spatial gradient of the input signal. Consequently, assuming the motion displacement $\delta(t)$ is sufficiently small, we approx-

$$\tilde{I}(x, t) \approx f(x + (1 + \alpha)\delta(t)) \tag{6}$$

quation (6) mathematically demonstrates that the subtle temporal displacement within the video sequence has been effectively amplified to $(1 + \alpha)\delta(t)$, thereby ving a net motion magnification of α.

or smooth, low-frequency spatial profiles (e.g., a low-frequency cosine wave) rgoing sub-pixel translations, this first-order Taylor approximation provides a y accurate representation of the shifted signal. However, for rapidly varying image ions $f(x)$ characterized by high spatial frequencies (e.g., sharp edges or fine res), the validity of the first-order approximation rapidly degrades. Under such tions, larger effective perturbation values, $(1 + \alpha)\delta(t)$—which scale aggressively both higher magnification factors α and larger base motions $\delta(t)$—introduce e unmodeled higher-order Taylor residuals, leading to prominent clipping artifacts noise amplification in the reconstructed video.

he aforementioned assumptions, while idealized and premised on minimal motion noise-free conditions, often diverge significantly from real-world scenarios. In practical applications, the subtle variations we anticipate can be substantial, resulting table deviations between the first-order Taylor expansion signal and the actual

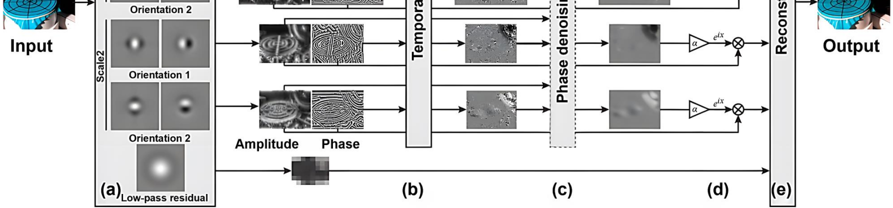


Fig. 7 Overview of the phase-based Eulerian video magnification framework[26]. The video decomposition process employs a complex steerable pyramid, facilitating the separation of local wavelet amplitude from phase information (a). Temporal filtering of the phase component is subsequently performed independently at each spatial position, orientation, and scale (b). Alternatively, amplitude-weighted spatial smoothing (c) can be applied to enhance phase SNR, thereby improving the final results. The temporally band passed phase is then selectively amplified or attenuated by an α-factor (d), followed by video reconstruction (e). This processing pipeline is demonstrated using a membrane sequence, with the pyramid decomposition limited to two scales and two orientations for visualization clarity.

To ensure theoretical consistency, the analysis is conducted under a one-dimensional (1D) local translational motion model. Conducting a Fourier series decomposition on the translated profile $f(x + \delta(t))$ yields a representation consisting of an infinite sum of complex sinusoids:

by completing the inverse pyramid synthesis.

n video signal processing, noise presents a formidable challenge, particularly due disruptive impact on intensity and amplitude profiles. Random noise fluctuations mask subtle, low-amplitude motion signals, severely undermining measurement racy. In contrast, local phase information exhibits remarkable geometric stability esilience against noise perturbations, making it an inherently more reliable feature notion manipulation [73]. Consequently, PEVM overcomes the noise sensitivity raints that limit LEVM, allowing for a significantly higher maximum allowable ification factor while preserving structural fidelity.

o theoretically substantiate the superior noise robustness of PEVM over its r counterpart, consider a comparative noise analysis within a local spatial sub-. Let the localized video signal be represented as a complex analytic signal $t) = A(x,t)e^{i\phi(x,t)}$, where $A(x,t)$ and $\phi(x,t)$ denote the local amplitude and e, respectively. Under an Additive White Gaussian Noise (AWGN) model, the -contaminated signal is expressed as $\tilde{S}(x,t) = S(x,t) + n(x,t)$, where $n \sim$ $\sigma^2)$. In prominent structural regions characterized by a high signal-to-noise ratio $\gg \sigma)$, the noise predominantly perturbs the amplitude envelope, while the induced e variance is tightly bounded by σ^2/A^2. Because LEVM directly scales the spa-ntensity gradients, it uniformly amplifies the additive noise variance by a factor

Limitations notwithstanding, these Eulerian operational profiles remain significantly more efficient than Lagrangian methods, which incur a polynomial complexity ranging from $\mathcal{O}(N \log N)$ to $\mathcal{O}(N^2)$ to compute dense optical flow fields. Consequently, while Lagrangian tracking offers fine-grained trajectory precision, its poor scalability renders it prohibitive for long-duration or high-definition monitoring, leaving Eulerian frameworks as the dominant paradigm for real-time structural health monitoring and biomedical diagnostics. To this day, PEVM remains a foundational benchmark that continues to inspire advanced algorithmic optimizations [31, 43, 46–48, 74, 75].

2.3 Deep Learning based Methods

2.3.1 Dataset Configurations and Performance Indicators

Deep learning methods generally require substantial training data to capture complex features and patterns [76, 77]. However, in minor motion magnification applications, collecting real-world sequences accompanied by precise physical reference remains an arduous and resource-intensive endeavor. Since motion magnification involves subtle, sub-pixel variations, data collection in real environments is often highly sensitive to noise, lighting fluctuations, and scene complexity, which can severely affect data quality.

32, 33] Swing, Gunshot).

undational Synthetic Configurations (Primary Supervised Training): e to the practical challenges in real-world labeling—such as the need for spelized high-speed equipment and sub-pixel annotation—researchers frequently erage virtual environments. A pioneering example introduced by Oh et al. [33] thesized training pairs using images from MS COCO [79] as backgrounds and gmented objects from PASCAL VOC [80] as foregrounds, as exemplified by the presentative configurations illustrated in Figure 8. In this configuration, random o-pixel 2D warping fields are applied to simulate non-rigid foreground movements, oviding explicit reference fields beneficial for baseline network convergence.

rectionally Decoupled Synthetic Configurations (Axial Training): AxiMM [78] constructs a synthetic training dataset using a copy-paste based pipeline, ere foreground objects from standard segmentation datasets are assigned transion vectors decomposed along a specified axis. This allows the generation of red samples with directionally constrained motion for supervised learning of axial tion magnification.

gh-Resolution and Perturbed Synthetic Configurations (Robustness

CAL VOC and background from MS COCO dataset. (b) Only has background moving to ensure the network learns global motion (c) Background blurred to ensure low contrast texture is well-represented. And another two parts with the same specification as (b) and (c) with background motion removed so that the network learns the changes that are due to noise only.The dataset is available at this link.

Rather than separating these samples into isolated datasets, current literature frequently groups them based on their motion conditions into: (i) *Static Mode*, where subtle vibrations occur against a completely static backdrop (e.g., periodic respiratory movements in *Baby* or high-frequency string vibrations in *Guitar*); and (ii) *Dynamic Mode*, where target minor motions are heavily contaminated by severe, large-scale concurrent macro-motions (e.g., facial micro-expressions accompanied by rigid head tilting in *Face* or gun barrel vibrations buried under hand recoil in *Gunshot*). These sequences primarily evaluate cross-domain generalization, structural stability, and resistance to edge bleeding. Notably, self-supervised paradigms like FlowMag [81] directly bypass synthetic dependency by executing test-time adaptation straight on these real-world sequences.

Multidimensional Evaluation Metrics for MM

ations	IQA [illegible], MANIQA		noise, shadows, and illumination changes.
putational Efficy	Inference Latency, FPS, FLOPs, Param Count	Reference-free	Benchmarks deployment viability for edge computing and real-time processing.

e $I(i,j)$ and $K(i,j)$ represent the pixel intensities at position (i,j), and m and the width and height. Based on MSE, the Peak Signal-to-Noise Ratio (PSNR) evaluates the signal strength relative to noise:

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{MAX^2}{\text{MSE}}\right), \tag{12}$$

e MAX is the maximum possible pixel value (typically 255 for 8-bit images).

owever, since MSE and PSNR assume independent pixel distributions and fail to unt for the human visual system's sensitivity to structural variations, the Struc- Similarity Index Measure (SSIM) [83] is preferred, incorporating luminance, ast, and structural information:

bandpass filtering, whereas deep neural networks optimize this dimension by integrating dynamic filters or inter-frame shape difference regularizations to penalize erratic phase variations across consecutive frames.

(d) Frequency Fidelity: Crucial for engineering diagnostics and health monitoring, frequency fidelity verifies whether the dominant frequency component of the amplified motion field precisely matches the source vibration spectrum without introducing spectral aliasing or artificial harmonics. This is typically evaluated by mapping pixel displacements over time into the frequency domain via Fast Fourier Transforms (FFT) and validating the resulting Power Spectral Density (PSD) peaks against baseline reference sensors [23].

(e) Robustness to Environmental Perturbations: Real-world deployments introduce non-ideal perturbations, making models highly sensitive to environmental factors. This indicator evaluates two main sub-dimensions: (i) *Noise Sensitivity*, which examines how a model handles camera photon noise to avoid amplifying high-frequency sensor noise alongside the true minor motion; and (ii) *Illumination Robustness*, which ensures that local shadow changes, light reflections, or global luminance transitions are not mistakenly captured as structural movements and erroneously magnified. Since traditional full-reference metrics fail when reference frames are missing in practical applications, No-Reference (NR) or Blind Image Quality Assessment (BIQA) frameain Focused Magnification (SDFM) and Multi-Domain Synergistic Magnification SM). Here, the term "domain" refers to either (1) a mathematical representation (e.g., spatial, frequency, or phase) or (2) a sensing modality (e.g., RGB frames ent streams).

nition 1 (Single-Domain Focused Magnification (SDFM)) An architecture is catego- as SDFM if feature extraction, motion manipulation, and reconstruction are performed n a single representation domain without explicit transformation into alternative ins. In practice, this typically corresponds to spatial-domain processing or standard feature spaces learned by convolutional or transformer-based networks.

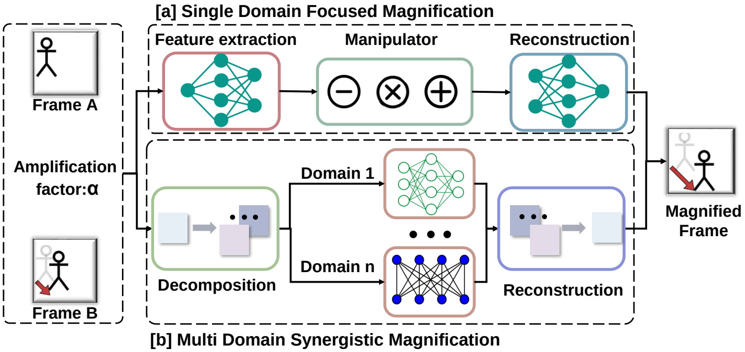
ormally, given an input sequence $I(\mathbf{x}, t)$, the magnified representation is obtained as $= \mathcal{M}(\mathcal{E}(I(\mathbf{x}, t)), \alpha)$, where the encoder $\mathcal{E}(\cdot)$ and manipulation operator $\mathcal{M}(\cdot)$ operate n the same representation space, and no explicit projection into frequency, phase, or modal domains is introduced.

nition 2 (Multi-Domain Synergistic Magnification (MDSM)) An architecture is cat- ed as MDSM if it explicitly transforms input representations into two or more ct domains—either mathematical (e.g., spatial-frequency or phase-amplitude) or physi- .g., frame-event modalities)—and performs motion amplification through cross-domain

Table 3 **Refined taxonomy and hardware-software configuration matrix of prominent learning-based motion magnification methodologies.**

Method	Input Type	Core Architecture	Loss Functions	Primary Datasets	Real-Time (FPS)	Class.
LB-VMM [33]	Consecutive RGB pairs	Fully Convolutional CNN	L_1	Spatial synthetic video set	Yes (∼60 FPS)	SDFM
LBA-VMM [35]	Consecutive RGB pairs	Axial-Constrained CNN	$L_1 + L_{\text{per}} + L_{\text{lap}}$	Axial vibration synthetic set	No (∼15 FPS)	SDFM
LN-VMM [38]	Continuous RGB frames	Feature-Sharing Lightweight CNN	$L_1 + L_e + L_p$	Lightweight spatial dataset	Yes (∼90 FPS)	SDFM
STB-VMM [36]	Consecutive RGB pairs	Hierarchical Swin Transformer	L_1	High-quality spatial synthetic set	No (∼8 FPS)	SDFM
MD-VMM [37]	Continuous RGB sequences	Spatial-Frequency Dual-Stream Net	$L_1 + L_e + L_p$	Frequency-decoupled synthetic pairs	No (∼5 FPS)	MDSM
FD-VMM [41]	Continuous RGB frames	Multi-Level Isomorphic Freq Decoupling Net	$L_1 + L_e + L_c$	Frequency-motion synthetic set	No (∼12 FPS)	MDSM
EB-VMM [42]	RGB frames + Event streams	Heterogeneous Cross-Modal Flow Net	$L_1 + L_{rc}$	Multimodal Ev-Magnify dataset	No (∼3 FPS)	MDSM

y: LN-VMM [38] and LB-VMM [33] deliver real-time processing capabilities with omputational overhead due to their streamlined CNN structures. In contrast, win-Transformer backbone of STB-VMM [36] and the forward/inverse spectral forms of MD-VMM [37] demand heavy GPU memory allocation and suffer from eal-time inference latency, presenting a critical trade-off between amplification ty and processing speed.



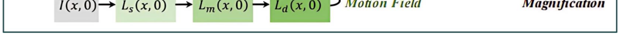


Fig. 10 The network framework of some classical deep learning methods: (a) shows the STB-VMM's[36] network; (b) shows the LB-VMM's[33] network; (c) shows the MD-VMM's[37] network; (d) shows the FD-VMM's[41] network.

et al. [26], expressed via the following objective relationship:

$$\hat{I}(x,t) = f(x + (1+\alpha)\delta(t)), \tag{15}$$

where the ultimate goal is to optimize a set of neural filters capable of implicitly isolating, manipulating, and amplifying the sub-pixel motion signal $\delta(t)$ while preserving the invariant static appearance features.

As visualized in the representative layout in Figure 10(b), the LB-VMM architecture decomposes the computational pipeline into three core functional modules: the encoder $G_e(\cdot)$, the manipulator $G_m(\cdot)$, and the decoder $G_d(\cdot)$. The network ingests two consecutive frames (X_a, X_b) along with the target magnification factor α. The encoder is structurally designed to disentangle the input stream into independent components, yielding texture features $V = G_{e,\text{texture}}(X)$ and shape features $M = G_{e,\text{shape}}(X)$. The manipulator subsequently operates exclusively within the shape feature space, mapurfaces characteristic of adversarial and perceptual objectives tend to destabi-ub-pixel feature manipulation, frequently introducing high-frequency geometric cts rather than enforcing strict structural fidelity. To enforce the explicit latent ation between texture and shape representations, a specialized perturbation egy is applied during training: the intensity of selected frames is randomly per-ed, forcing the network to minimize texture representation error while keeping orresponding shape vectors invariant.

he operational execution of LB-VMM is split into two modes: static mode and mic mode. The static mode utilizes the initial video frame as a constant structural or, processing the frame pair (X_1, X_t), which strictly adheres to the baseline ian assumption. In contrast, the dynamic mode calculates variations between ent frames, processing the pairs (X_{t-1}, X_t) to amplify localized frame-to-frame ity fields, changing the physical interpretation of α. Furthermore, a temporal ng module was introduced within the manipulator to isolate specific frequency s. However, subsequent evaluations confirmed that because the fully convolutional sentation inherently possesses stable noise-reconstruction traits, explicit temporal ng is largely redundant for basic spatial stability [36]. From a critical standpoint, onvolutional network design of LB-VMM relies on the fundamental mathematical nption of spatial local correlation and translation invariance, processing features processing (∼90 FPS). The foundational assumption of LN-VMM is that spatial structure boundaries can be explicitly regulated via surrogate boundary constraints. Although the network maintains exceptionally sharp object edges due to the specialized edge loss, its compact parameter capacity bounds its latent representation capability. As a result, its characteristic failure mode manifests under large magnification factors ($\alpha > 10$) or out-of-distribution noise, where the lightweight layers undergo an information bottleneck, triggering severe structural distortion and texture erasing.

Based on the long-range relational modeling capabilities of vision transformers [91], Lado-Roigé et al. [36] presented STB-VMM, substituting the standard convolutional encoder with a hierarchical Swin Transformer backbone [92]. Utilizing a shifted-window self-attention mechanism, STB-VMM exhibits outstanding resilience to heavy sensor noise and successfully suppresses the frame-to-frame temporal flicker that typically plagues pure CNN architectures. The core mathematical assumption of STB-VMM is that sub-pixel motion details are non-locally correlated across broader spatial regions and can be optimally isolated through global relational matrices. This token aggregation over localized windows serves as an implicit spatial-temporal smoother, enabling the network to excel at noise suppression and preserve sharp localized boundaries. However, the critical trade-off of this Transformer architecture is its quadratic computational complexity ($O(N^2)$) relative to token length, which induces non-real-

3 Multi-Domain Synergistic Magnification

leviate the limitations of single-domain representations, recent deep learning-l methods increasingly explore multi-domain strategies for motion disentangle- and amplification. As defined in Section 2.X, Multi-Domain Synergistic Mag-tion (MDSM) explicitly introduces transformations into multiple representation ins, enabling complementary modeling of motion and appearance.

a general formulation, the input sequence $I(\mathbf{x}, t)$ is projected into a set of in-specific subspaces $\mathcal{Z} = \{\mathcal{Z}_1, \mathcal{Z}_2, \ldots, \mathcal{Z}_n\}$, where different components capture ct aspects of the signal (e.g., motion, texture, or structural information). Motion ification is then selectively applied to motion-relevant components, followed by -domain fusion to reconstruct the output $\hat{I}(\mathbf{x}, t) = \mathcal{D}(\mathcal{Z}_{\text{mag}})$.

representative example of mathematical cross-domain transformation (Rule 2) is VMM [37], which operates in the frequency domain. The model decomposes input es into amplitude and phase components via Fourier transform, where motion is oximately encoded as phase variations under small motion assumptions. Ampli-on is applied to the phase component, while amplitude is largely preserved to tain structural consistency. This strategy improves robustness to noise compared motion scenarios.

Overall, MDSM methods demonstrate improved robustness in motion disentanglement and noise suppression compared to single-domain approaches. However, these advantages often come at the cost of increased model complexity, higher computational requirements, and additional sensitivity to domain-specific assumptions. Designing efficient and robust cross-domain representations therefore remains an open challenge.

2.4 Critical and Comparative Synthesis of Methodologies

A cross-paradigm synthesis of motion magnification (MM) methodologies reveals an inherent trade-off among mathematical interpretability, computational efficiency, and representation fidelity. To systematically characterize the current landscape, these approaches can be examined through their underlying kinematic assumptions and typical failure modes.

Lagrangian methods are based on the explicit modeling of *spatial trajectory tracking*. In principle, they provide accurate displacement estimation through dense motion fields, which is advantageous for structural analysis. However, their computational cost scales significantly with the complexity of optical flow or keypoint matching, limiting practical deployment. In addition, their performance may degrade in the pres-

lerian Methods are well-suited for resource-constrained environments and peri-
c motion analysis, where computational efficiency and stable temporal filtering
required.

OFM approaches provide a balance between efficiency and reconstruction quality,
king them applicable to real-time scenarios with moderate motion complexity.

DSM methods are more suitable for complex scenes involving low signal-to-noise
ios, non-rigid motion, or multi-modal inputs, where cross-domain representations
n improve motion disentanglement.

Applications

achine vision tasks continue to increase in complexity, driven by the massive
ne and diversity of modern video data, MM has evolved from a visualization
ique into a functional component within broader analytical pipelines. Rather
serving as a standalone tool, MM is increasingly integrated to enhance subtle
o-temporal variations and improve the interpretability of downstream tasks. An
iew of representative application domains is illustrated in Figure 11.

eploying MM in real-world pipelines requires balancing data characteristics,
on scale, and validation protocols. Existing studies indicate that application

Fig. 11 Empirical taxonomy of motion magnification workflows across diverse analytical fields, featuring representative implementations in visual vibration measurement [44, 94, 95], structural health monitoring [48, 96, 97], biomedical science [98–100], deepfake detection [101–103], and micro-expression recognition [104–106].

Validation is commonly performed by comparing magnified motion with synchronized measurements from Laser Doppler Vibrometers (LDV) [44, 97] or motor-based sensing systems [96]. Performance is evaluated using Power Spectral Density (PSD), Frequency Response Functions (FRF), modal identification [8], and Fourier-domain error metrics [44].

In practice, high-frequency camera jitter may be unintentionally amplified, while specular reflections on metallic surfaces can introduce phase estimation errors, potentially degrading reconstruction accuracy [96].

3.2 Structural Health Monitoring

In structural health monitoring (SHM), MM is applied to large-scale infrastructure such as bridges, buildings, and wind turbines [111–115]. Data are typically

ntion	200 fps)	facial muscle motion	tions	score	occlusion
ke Detec-	Compressed video (30 fps)	Subtle temporal inconsistencies	Cross-dataset testing	AUC-ROC, error rate	Compression artifacts

Biomedical Science

edical applications use MM as a non-invasive tool for monitoring physiological ls and assisting in microsurgical analysis [120]. Input data typically consist of or Near-Infrared (NIR) video capturing subtle skin color variations or chest on [101, 121]. The target signals include cardiovascular pulse and respiratory-ced displacement.

alidation is conducted using synchronized clinical devices such as Photoplethys-aphy (PPG), Electrocardiograms (ECG), and respiratory belts [101, 121]. rmance is evaluated using heart rate (BPM) error, respiratory rate (RR), and l-to-noise ratio (SNR).

owever, large body motion can disrupt motion continuity, and variations in skin and illumination may attenuate physiological signals, often requiring additional using metrics such as AUC-ROC, Equal Error Rate (EER), and classification accuracy [136, 137].

A major limitation arises from lossy compression, which suppresses high-frequency motion signals and may introduce artificial patterns. In such cases, magnification can amplify compression artifacts or temporal flicker instead of meaningful motion, potentially affecting downstream feature extraction methods such as LBP, HOOF, or CNN-LSTM pipelines [103, 104, 136]. Consequently, learning-based magnification frameworks often require continual adaptation to maintain robustness against evolving deepfake generation techniques [138].

4 Challenges and Future Directions

Despite significant progress in domain-specific deployments, extending MM from controlled experimental settings to unconstrained real-world environments reveals several persistent theoretical and engineering limitations. These challenges can be broadly categorized into three interrelated aspects: physical signal disentanglement, data representation constraints, and computational scalability.

4.1 Physical Disentanglement and Physics-Informed

ated through simplified transformations, such as linear 2D warping of static es. While effective for learning basic motion representations, these datasets fail pture the complexity of real-world conditions, including non-rigid deformation, re variability, illumination changes, and compression artifacts.

s a result, models trained exclusively on synthetic data often exhibit perfore degradation when applied to real-world scenarios. Addressing this issue requires oving both data diversity and domain generalization.

ne promising direction is unsupervised domain adaptation, where latent feature butions are aligned across synthetic and real domains using techniques such lversarial learning or distribution discrepancy minimization. These approaches urage models to extract motion features that are robust to environmental tions.

nother direction involves generative data augmentation. Recent advances in video sion models enable the synthesis of realistic minor motion sequences by condig on physical motion priors. Such generative frameworks can produce data with r textures, realistic noise characteristics, and more complex motion patterns, by improving training coverage and generalization.

Hardware-Aware Compression and Edge Scaling

unified mathematical formulation that abstracts diverse approaches under a common reconstruction objective, and proposes a two-tier taxonomy that categorizes deep learning-based methods into Single-Domain Focused Magnification (SDFM) and Multi-Domain Synergistic Magnification (MDSM). In addition, a systematic organization of application domains, datasets, and evaluation protocols is provided to support reproducible comparison and informed method selection.

Based on this structured perspective, several consistent patterns emerge across existing methodologies. Classical Lagrangian and Eulerian approaches offer strong interpretability and computational efficiency, but remain limited in handling complex, non-rigid motion and large amplification factors. In contrast, learning-based methods significantly enhance robustness and flexibility through data-driven representation learning, yet introduce challenges related to computational cost, generalization, and physical interpretability. Furthermore, the analysis across multiple application domains reveals that motion magnification is inherently context-dependent, with performance strongly influenced by domain-specific factors such as motion scale, signal-to-noise ratio, sensing conditions, and validation constraints.

Looking forward, advancing minor motion magnification requires a more integrated perspective that bridges physical modeling, data-driven learning, and system-level optimization. Future progress is likely to depend on improving physical consistency and interpretability through physics-informed constraints, enhancing robustness to

ducibility at the procedural level.
o new code, pretrained models, or curated datasets were generated in this study. re work will aim to further enhance openness by releasing structured benchmarks, lardized evaluation protocols, and curated resources where feasible.

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clarations

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